

## Assessment Test: Software Tester (QA)

|                     |                                     |
|---------------------|-------------------------------------|
| Duration: 01:15 Hrs |                                     |
|                     |                                     |
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# Digitize Your Business

You are testing a **Task Management Application** with the following features: ● User **Registration & Login**

- Logged-in users can:
  - Create a task
  - View task list
  - Edit a task
  - Delete a task
- Tasks are stored in a database and displayed in a list

Assume the application is **close to production release** and has **no existing test documentation**.

### Test Case Design

Write **test scenarios and test cases** for:

- Registration
- Login
- Task CRUD operations
- Input validation & error handling

Include **positive, negative, and edge cases**.

# Task Management Application – Software Testing Assignment

## 1. Objective

The objective of this testing exercise is to verify that the Task Management Application works correctly, handles invalid inputs properly, protects user data, and is reliable enough for a production release.

The application provides the following functionality:

- User Registration
- User Login
- Create Task
- View Task List
- Edit Task
- Delete Task
- Database storage of tasks

The testing approach covers positive, negative, and edge cases, along with validation, error handling, security, and data integrity.

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## 2. Assumptions

Since detailed requirements and existing test documentation are not provided, the following assumptions are made:

- Email is used as the unique identifier for user login.
  - Required registration and task fields should not accept empty values.
  - A user should only be able to access and manage their own tasks.
  - Tasks should be permanently stored in the database.
  - The application should display meaningful error messages for invalid input or system failures.
  - Exact field-length limits and password rules should follow the application's defined requirements.
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## 3. Test Case Design

### 3.1 Registration Test Cases

| TC ID  | Test Scenario                          | Test Data / Steps                                                                     | Expected Result                                      | Type     |
|--------|----------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------|----------|
| REG-01 | Register with valid details            | Enter valid name, email and password and submit                                       | Account should be created successfully               | Positive |
| REG-02 | Register with another valid email      | Enter valid details with a new email                                                  | New account should be created                        | Positive |
| REG-03 | Register with already registered email | Use an email that already exists                                                      | Registration should fail with an appropriate message | Negative |
| REG-04 | Register with invalid email            | Enter values such as <code>abc</code> , <code>test@</code> , or <code>test.com</code> | Application should display invalid email validation  | Negative |

|        |                                            |                                                               |                                                                       |          |
|--------|--------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------|----------|
| REG-05 | Leave required fields empty                | Leave name, email or password blank                           | Registration should not proceed and validation messages should appear | Negative |
| REG-06 | Enter invalid/weak password                | Enter a password that does not satisfy defined password rules | Application should reject it and display an appropriate message       | Negative |
| REG-07 | Test minimum password length               | Enter password at the minimum allowed length                  | Registration should succeed if all other details are valid            | Edge     |
| REG-08 | Test password exceeding maximum length     | Enter a password longer than the defined limit                | Application should reject or properly handle the input                | Edge     |
| REG-09 | Enter spaces around email                  | Enter email with leading/trailing spaces                      | Application should handle whitespace correctly                        | Edge     |
| REG-10 | Enter special characters in allowed fields | Use valid special characters where permitted                  | Application should process input correctly without errors             | Edge     |

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## 3.2 Login Test Cases

| TC ID    | Test Scenario                     | Test Data / Steps                               | Expected Result                                                                    | Type              |
|----------|-----------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------|-------------------|
| LOGIN-01 | Login with valid credentials      | Enter registered email and correct password     | User should successfully log in and access the task dashboard                      | Positive          |
| LOGIN-02 | Login with incorrect password     | Enter registered email with wrong password      | Login should fail with an appropriate error message                                | Negative          |
| LOGIN-03 | Login with unregistered email     | Enter an email that does not exist              | Login should fail                                                                  | Negative          |
| LOGIN-04 | Login with empty fields           | Leave email and/or password empty               | Validation message should be displayed                                             | Negative          |
| LOGIN-05 | Login with invalid email format   | Enter an invalid email format                   | Application should display email validation error                                  | Negative          |
| LOGIN-06 | Password should be masked         | Enter password in the password field            | Password should not be displayed as plain text                                     | Positive          |
| LOGIN-07 | Multiple incorrect login attempts | Enter incorrect credentials repeatedly          | Application should handle repeated failures according to its security requirements | Edge              |
| LOGIN-08 | Logout and access protected page  | Login, logout and try to access task page again | User should not be able to access protected task data                              | Negative/Security |

# 3.3 Task CRUD Test Cases

CRUD means Create, Read, Update and Delete.

## A. Create Task

| TC ID     | Test Scenario                      | Test Data / Steps                                        | Expected Result                                          | Type     |
|-----------|------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------|
| TASK-C-01 | Create task with valid details     | Enter valid task title/description and save              | Task should be created successfully                      | Positive |
| TASK-C-02 | Create multiple tasks              | Create several valid tasks                               | All tasks should be created and displayed correctly      | Positive |
| TASK-C-03 | Create task without title          | Leave mandatory title empty and submit                   | Task should not be created and validation should appear  | Negative |
| TASK-C-04 | Create task with empty description | Leave description empty if it is optional                | Task should be created successfully                      | Edge     |
| TASK-C-05 | Enter maximum allowed title length | Enter title at defined maximum length                    | Task should be created if within allowed limit           | Edge     |
| TASK-C-06 | Exceed maximum title length        | Enter title beyond defined limit                         | Application should reject or properly validate the input | Edge     |
| TASK-C-07 | Enter special characters           | Enter valid special characters in task title/description | Application should handle them correctly and safely      | Edge     |

## B. View Task List

| TC ID     | Test Scenario                | Test Data / Steps             | Expected Result                                                   | Type              |
|-----------|------------------------------|-------------------------------|-------------------------------------------------------------------|-------------------|
| TASK-R-01 | View existing tasks          | Login and open task list      | User should see their saved tasks                                 | Positive          |
| TASK-R-02 | View task list with no tasks | Login as a user with no tasks | Appropriate empty-state message should be displayed               | Edge              |
| TASK-R-03 | Create task and refresh page | Create task, refresh browser  | Task should still be present because it is stored in the database | Positive          |
| TASK-R-04 | Verify another user's tasks  | Login as a different user     | User should not see another user's tasks                          | Negative/Security |

|           |                            |                     |                                                                                          |      |
|-----------|----------------------------|---------------------|------------------------------------------------------------------------------------------|------|
| TASK-R-05 | View large number of tasks | User has many tasks | Application should display tasks correctly without data loss or major performance issues | Edge |
|-----------|----------------------------|---------------------|------------------------------------------------------------------------------------------|------|

## C. Edit Task

| TC ID     | Test Scenario                 | Test Data / Steps                         | Expected Result                                      | Type              |
|-----------|-------------------------------|-------------------------------------------|------------------------------------------------------|-------------------|
| TASK-U-01 | Edit existing task            | Change task title/description and save    | Updated task should be displayed correctly           | Positive          |
| TASK-U-02 | Edit task with valid new data | Replace existing data with valid values   | Changes should be saved successfully                 | Positive          |
| TASK-U-03 | Remove mandatory task title   | Clear title and save                      | Update should fail with validation message           | Negative          |
| TASK-U-04 | Cancel editing                | Open edit option, make changes and cancel | Original task data should remain unchanged           | Edge              |
| TASK-U-05 | Edit another user's task      | Attempt to modify another user's task     | Application should prevent unauthorized modification | Negative/Security |
| TASK-U-06 | Refresh after editing         | Edit task and refresh page                | Updated data should remain available                 | Positive          |

## D. Delete Task

| TC ID     | Test Scenario                            | Test Data / Steps                        | Expected Result                                                                                   | Type              |
|-----------|------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------|
| TASK-D-01 | Delete existing task                     | Select a task and delete it              | Task should be removed from the task list and database                                            | Positive          |
| TASK-D-02 | Cancel deletion                          | Select delete and cancel confirmation    | Task should remain unchanged                                                                      | Edge              |
| TASK-D-03 | Delete another user's task               | Attempt to delete another user's task    | Application should prevent unauthorized deletion                                                  | Negative/Security |
| TASK-D-04 | Verify deletion after refresh            | Delete a task and refresh the page       | Deleted task should not reappear                                                                  | Positive          |
| TASK-D-05 | Delete task when database is unavailable | Attempt deletion during database failure | Application should display an appropriate error and should not falsely report successful deletion | Negative          |

# 3.4 Input Validation and Error Handling

| TC ID  | Test Scenario                      | Test Data / Steps                                          | Expected Result                                                                        | Type     |
|--------|------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------|----------|
| VAL-01 | Submit empty required fields       | Leave required fields blank                                | Validation messages should be displayed                                                | Negative |
| VAL-02 | Enter invalid email                | Enter malformed email                                      | Application should reject invalid email                                                | Negative |
| VAL-03 | Enter extremely long input         | Enter input exceeding allowed length                       | Application should reject or safely handle it                                          | Edge     |
| VAL-04 | Enter special characters           | Enter characters such as @#\$%^&* where applicable         | Application should safely process valid input                                          | Edge     |
| VAL-05 | Enter HTML/script input            | Enter potentially malicious input such as script tags      | Application should sanitize/reject unsafe input and should not execute it              | Security |
| VAL-06 | Database unavailable               | Perform create/update/delete while database is unavailable | User should receive a meaningful error message instead of application crash            | Negative |
| VAL-07 | Server error                       | Trigger/encounter an unexpected server error               | Application should display a user-friendly error instead of exposing stack traces      | Negative |
| VAL-08 | Network interruption               | Interrupt network during task submission                   | Application should handle the failure gracefully and avoid misleading success messages | Edge     |
| VAL-09 | Refresh after successful operation | Create/edit/delete a task and refresh                      | Database state and UI should remain consistent                                         | Positive |
| VAL-10 | Invalid/expired session            | Attempt to perform an operation after session expiry       | User should be redirected to login or asked to authenticate again                      | Security |

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### Bug Identification

List **8–10 potential bugs or risk areas** without executing the application. For each bug, mention:

- Bug description
- Severity (Critical / Major / Minor)
- Reason/Impact

# Bug Identification – Potential Bugs / Risk Areas

The following potential bugs and risk areas are identified based on the application's requirements and without executing the application. These are areas that should be specifically verified during testing before production release.

| Bug ID | Bug Description                                                                                                                                                              | Severity | Reason / Impact                                                                                                                      |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| BUG-01 | <b>Duplicate user registration:</b> The application may allow a user to register multiple accounts using the same email address.                                             | Major    | Email is generally expected to be unique for an account. Duplicate accounts can cause login confusion and account-management issues. |
| BUG-02 | <b>Unauthorized access to another user's tasks:</b> A logged-in user may be able to view tasks belonging to another user by changing or manipulating a task/user ID.         | Critical | This can expose private user data and represents a serious authorization and data privacy issue.                                     |
| BUG-03 | <b>Unauthorized task modification/deletion:</b> A user may be able to edit or delete another user's task.                                                                    | Critical | This can result in unauthorized modification or permanent loss of another user's data.                                               |
| BUG-04 | <b>Passwords stored insecurely:</b> Passwords may be stored as plain text or using weak protection in the database.                                                          | Critical | If the database is compromised, user passwords could be exposed, creating a serious security risk.                                   |
| BUG-05 | <b>Task data not persisted correctly:</b> A newly created or updated task may appear in the UI but may not be correctly saved in the database.                               | Major    | Tasks could disappear after refreshing the page or logging in again, resulting in data loss and poor user experience.                |
| BUG-06 | <b>Insufficient input validation:</b> The application may accept empty, excessively long, or invalid values for registration and task fields.                                | Major    | Invalid data can enter the database and may cause application errors, poor data quality, or unexpected behavior.                     |
| BUG-07 | <b>SQL Injection vulnerability:</b> User input may be directly used in database queries without proper validation or parameterization.                                       | Critical | An attacker could manipulate database queries and potentially access, modify, or delete sensitive data.                              |
| BUG-08 | <b>Session/logout issue:</b> A user may still be able to access protected task information after logging out, for example through the browser's Back button or cached pages. | Major    | Unauthorized users could potentially access private task information after the session should have ended.                            |
| BUG-09 | <b>Incorrect task deletion:</b> Due to incorrect task ID handling, the application may delete a different task from the one selected by the user.                            | Major    | This could cause unintended permanent data loss and affect data integrity.                                                           |
| BUG-10 | <b>Poor error handling:</b> Database, network, or server failures may result in application crashes or expose technical details such as stack traces to the user.            | Major    | It can lead to a poor user experience and may expose sensitive implementation details that could create security risks.              |

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## 5. Testing Approach

Because the application is close to production release, I would prioritize testing in the following order:

### 1. Functional Testing

Verify that registration, login and all CRUD operations work according to requirements.

### 2. Validation Testing

Verify that empty, invalid, oversized and unexpected inputs are handled correctly.

### 3. Security Testing

Verify authentication, authorization, session handling, password security and protection against common attacks such as SQL injection.

### 4. Data Integrity Testing

Verify that data created or modified through the UI is correctly stored in the database and remains consistent after refresh or re-login.

### 5. Error Handling

Verify that database, network and server failures do not crash the application and that users receive meaningful error messages.

### 6. Edge Case Testing

Test boundary values, large numbers of tasks, unusual characters, repeated operations and unexpected user behavior.

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## 6. Conclusion

The testing strategy covers the major functional and non-functional risk areas of the Task Management Application.

Before production release, special attention should be given to authentication, authorization, database persistence, input validation, error handling and data security. The potential bugs listed above should be verified through actual testing, and any confirmed defects should be logged, prioritized and retested after fixes.

The objective is not only to verify that the application works under normal conditions, but also to ensure that it behaves safely and reliably when users provide invalid input, encounter system failures, or perform unexpected actions.

### Note

- You are **not expected to complete everything perfectly**.
- We evaluate **how you think, structure, and explain**, not just what you write.

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